

ALGEBRA 2 REVIEW

Name: _____

Date: _____ Per: _____

WEEK 1

SET C

1. Which statement is not true?

- A. Rational numbers are closed under division.
- B. Odd numbers are closed under multiplication.
- C. Prime numbers are closed under subtraction.
- D. Integers are closed under addition.

C

2. What is the value of p in the equation below?

$$-\frac{3}{2}\left(16 + \frac{4}{9}p\right) = \frac{1}{4}\left(\frac{1}{3}p - 6\right)$$

$$= -24 - \frac{2}{3}p = \frac{1}{12}p - \frac{3}{2}$$

$$-\frac{45}{2} = \frac{3}{4}p$$

$p = -30$

3. Approximate the value of the expression below to the nearest hundredth.

$$\frac{15\sqrt{2}-1}{8} \approx 2.52665$$

2.53

4. Find the value of the expression below if $a = -\frac{2}{3}$ and $b = \frac{1}{12}$. Give your answer as a fraction in simplest form.

$$\frac{15}{8}\left(-\frac{2}{3}\right)^2 + \left(-\frac{2}{3}\right)\left(\frac{1}{12}\right) \frac{15}{8}a^2 + ab$$

$$= \frac{15}{8}\left(\frac{4}{9}\right) - \frac{1}{18}$$

$$= \frac{5}{6} - \frac{1}{18} = \frac{14}{18}$$

$\frac{7}{9}$

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WEEK 1

SET D

1. Find the value of the expression below.

$$\frac{-8|-18+2(-3)|}{6+2\sqrt{9}} = \frac{-8|-18-6|}{6+6}$$

$$= \frac{-8|-24|}{12}$$

$$= -16$$

- A. 4
- B. -4
- C. 16
- D. -16

D

2. Which list contains values that are integers only?

- A. $\frac{1}{4}, \sqrt{20}, -11$
- B. $-\sqrt{8}, \sqrt{60}, \sqrt{100}$
- C. $-5\sqrt{81}, 17, -\frac{42}{3} = -45, 17, -14$
- D. $1.5, 36, \frac{28}{4}$

C

3. Given three consecutive odd integers, the sum of the smallest integer and twice the largest integer is ten squared more than the median integer. What is the largest integer?

$$x + 2(x+4) = x+2+10^2$$

$$3x+8 = x+102$$

$$2x = 94$$

$$x = 47$$

- A. 49
- B. 51
- C. 53
- D. 57

47, 49, 51

B

4. Which property is illustrated by the statement below?

$$(14 \cdot 2k) + k^2 = k^2 + (14 \cdot 2k)$$

- A. Distributive Property
- B. Commutative Property of Multiplication
- C. Associative Property of Addition
- D. Commutative Property of Addition

D

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